

Midterm Report

This report presents the result of the implementation of the “Deep Learning with Differential Privacy” paper by M. Abadi, et al. The main idea of the paper is to implement Differential Privacy-Stochastic Gradient Descent (DP-SGD) to a deep learning model to observe the trade off between privacy and utility. The model that is used is a CNN model on MNIST dataset.

Differential privacy was first introduced by Dwork et al in 2006. It is a mathematical framework that provides rigorous and quantifiable privacy guarantees. The main idea is the output of a function should be essentially the same whether or not an element is included in the input. It ensures that the adversary can not learn any meaningful information about any specific person by observing the model’s output or parameters. Formally, a randomized mechanism M satisfies (ϵ, δ) -differential privacy if for any two datasets D and D' differ by exactly one record, and for any possible output S :

$$Pr[M(D) \in S] \leq e^{\epsilon} \times Pr[M(D') \in S] + \delta$$

Where Epsilon is the privacy budget and Delta is the failure probability

DP-SGD modifies the standard mini batch SGD training process by introducing two key steps at each gradient update step:

1. Gradient Clipping

Rather than computing a single gradient over the entire batch, DP-SGD computes each element’s gradient and clips each one so that its L2 normalization does not exceed a fixed threshold C (clipping norm / sensitivity).

2. Gaussian Noise Addition

After clipping, the Gaussian noise is added to the sum of the clipped gradients before the model parameters are updated. The standard deviation of the noise is set to $\sigma = \text{noise_multiplier} \times C$. A larger noise multiplier adds more noise but lower model accuracy.

The dataset that is used in this experiment is the MNIST dataset which consists of 70,000 grayscale images of handwritten numbers (0 to 9), and divided to 60,000 training images and 10,000 test images. Each image has the resolution of 28x28 pixels with mean = 0.1307 and std = 0.3081.

The model architecture follows the architecture that is used in “Deep Learning with Differential Privacy” paper by Abadi et al. It consists of two CNN layers followed by two fully connected layers:

- Conv(16, 8x8, stride=2, padding=2) + ReLU + MaxPool(2x2)
- Conv(32, 4x4, stride=2, padding=0) + ReLU + MaxPool(2x2)
- Fully connected 32 -> 10 (output logits)

For the DP-SGD Experiment, the following Hyperparameter settings is used:

- Optimizer = DP-SGD (Opacus)
- Learning rate = 0.25
- Batch size = 256
- Number of epochs = 10
- Gradient clipping norm $\odot$ = 1.0
- Delta = 1e-5
- Noise multipliers = 0.5, 0.7, 1.0, 1.3

The baseline model (without DP) is trained using the same parameter except the learning rate used is 0.01 instead of 0.25.

The perturbation method consists of two steps that are applied during each gradient update:

1. Per Sample Gradient Clipping

Each sample gradient is computed individually and clipped so that their L2 norm does not exceed the clipping threshold. Formally, the formula is:

$$\tilde{g}_i = g_i / \max(1, \|g_i\|_2 / C)$$

It ensures that no single training sample can influence the model update by more than a fixed amount

2. Gaussian Noise Addition

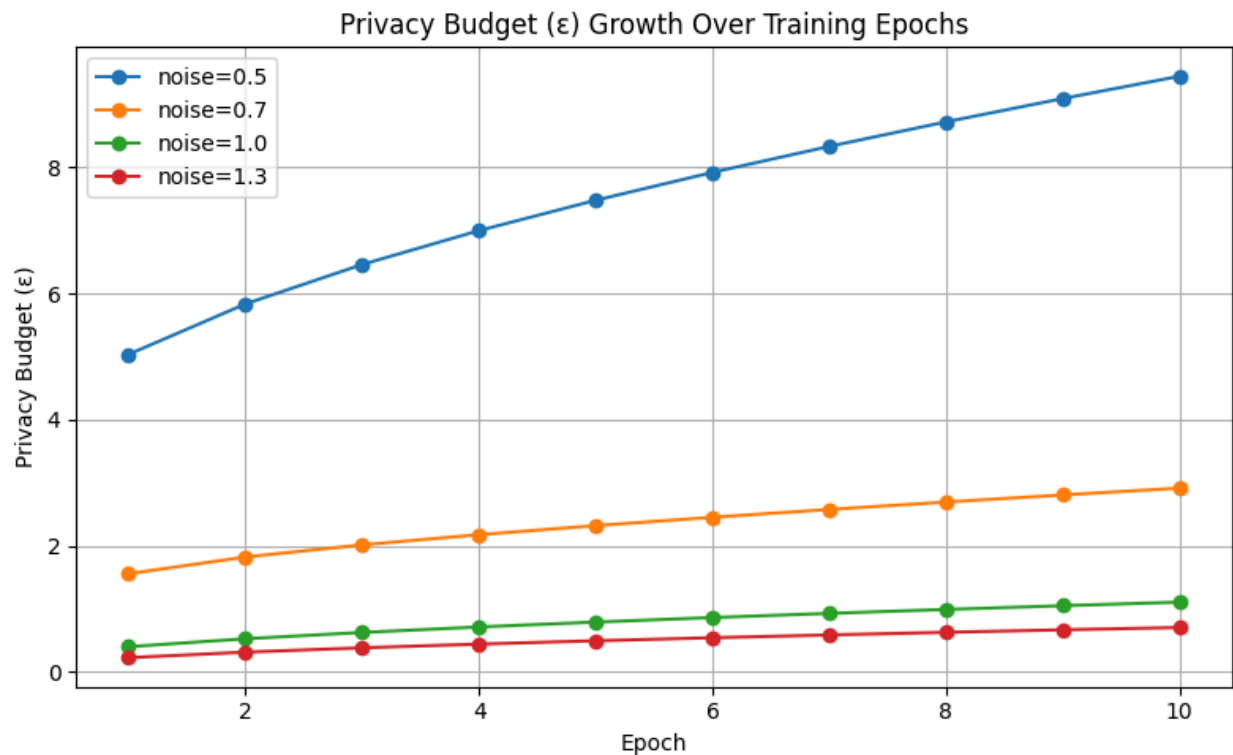
The noise is drawn from a Gaussian distribution with mean zero and std = noise_multiplier x C. The noisy gradient update is:

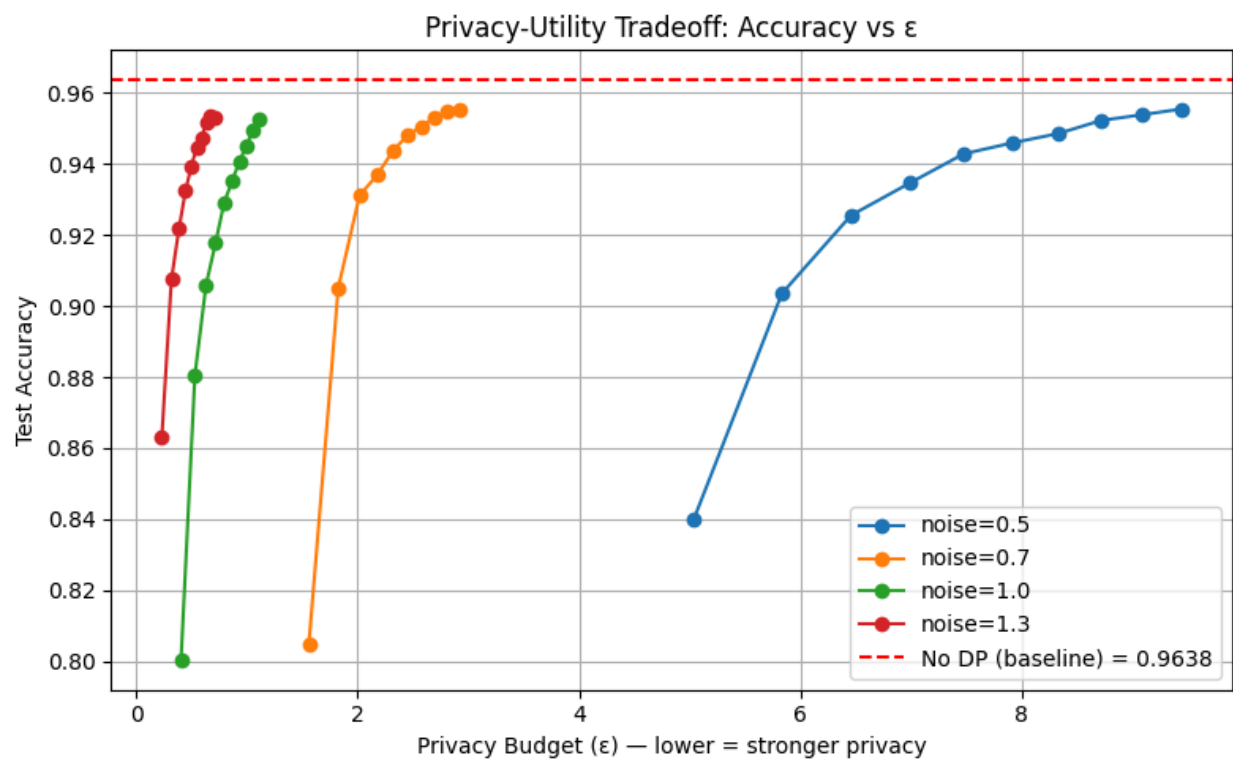
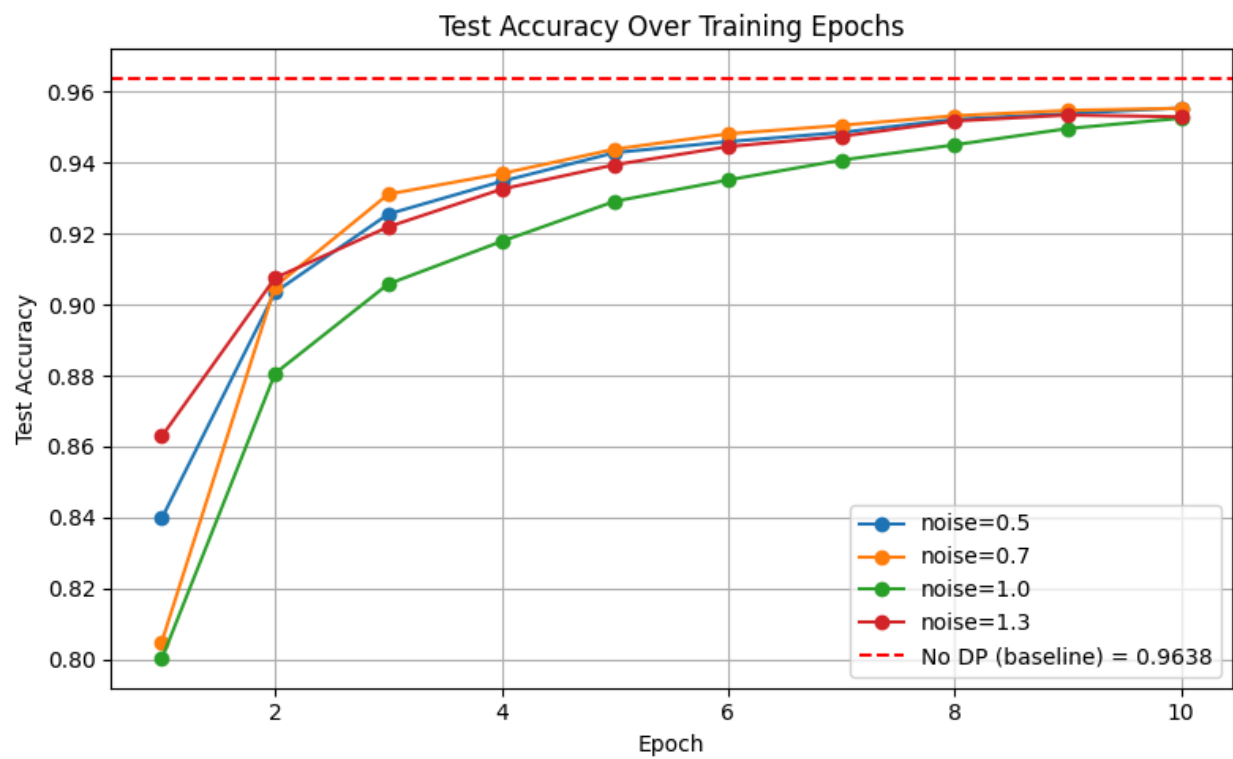
$$\tilde{g} = \sum_i \tilde{g}_i + N(0, \sigma^2 C^2 I)$$

The noise masks the contribution of individual samples, making it statistically impossible for an attacker to determine whether any particular person's data was included in the training set.

Here is the results:

Noise Multiplier	Final ϵ (privacy)	Test Accuracy (%)
0.5	9.44 (weakest privacy)	95.54
0.7	2.92	95.53
1.0	1.11	95.25
1.3	0.71 (strongest privacy)	95.29
Baseline	∞ (no privacy)	96.38





The baseline model has an accuracy of 96.38% on the test dataset of MNIST. Here are some observations from this experiment:

- The second highest performing model after the baseline model is the one with $\epsilon = 9.44$ with test accuracy of 95.54%. It seems that the weakest privacy model has the highest accuracy, so it proves the hypothesis that privacy and utility is a tradeoff.
- The lowest performing model is the most private model with $\epsilon = 0.71$ with test accuracy of 95.29%. It has 1.09 percentage points showing that DP-SGD has relatively small trade off.
- From the results we can see that ϵ is inversely proportional to accuracy. Showing that higher privacy will result in lower accuracy.